

his study develops a constructive material model for a subset of the statistical regularities of the Voynich Manuscript (VM). Rather than assuming a plaintext language, it asks whether low-order character structure can be implemented by a materially simple procedure without modern computation. Analysis of Herbal A shows that short-range character transitions retain substantial structural information: conditioning the next character on the preceding two-character context improves prediction, while low-order transition frequencies also vary systematically at the level of the physical bifolium.

These results motivate a physical state machine consisting of a 24×24 open-compartment shelf representing SPACE plus twenty-three VM character types. Each compartment stores a two-state context, while movable tablets specify a possible next character and a stopping threshold; tablet multiplicities implement local transition weights. Paragraph beginnings are handled by a separate Paragraph START box.

A concrete reconstruction of bifolium A-2 uses 600 main-shelf tablets and eight Paragraph START tablets. In simulation, the fitted apparatus closely reproduces A-2 word-length statistics and low-order transition distributions. Across 5,000 runs, 77.05% of generated tokens matched complete forms attested somewhere in the VM, including 22.05% whose attestation lay outside the A-2 fitting material. However, matched bigram and trigram null models show that high complete-word recurrence is largely a consequence of local transition structure rather than a shelf-specific effect. The more distinctive contribution of the apparatus is its finite material implementation of these constraints while maintaining a narrower VM-like word-length distribution and greater lexical diversity than corresponding unconstrained Markov generators.

Although this reconstruction does not constitute evidence that the VM was historically produced by this specific apparatus, it shows that a combinatorial mode of construction may provide a useful framework for investigating its surface structure.

Reproducing the Surface Structure of the Voynich Manuscript with a Material Ars Combinatoria: A Materially Implementable 24×24 Shelf-and-Tablet Reconstruction

Takashi Izumo^{1*}

^{1*}College of Law, Nihon University, Kandamisakicho, Chiyodaku, 1018375, Tokyo, Japan, ORCID: 0000-0003-0008-4729.

Corresponding author(s). E-mail(s): izumo.takashi@nihon-u.ac.jp;

Keywords: Voynich Manuscript; material combinatorics; n-gram structure; stochastic text generation; historical cryptology

1 Introduction

The Voynich Manuscript has commonly been approached as an undeciphered linguistic or cryptographic text. A major recent synthesis by Bowerman and Lindemann surveys and evaluates linguistic approaches to the manuscript and provides a useful overview of the statistical and linguistic questions that frame current research [1]. The present study does not attempt to disprove such interpretations. Instead, it asks a narrower constructive question: how much of the manuscript’s statistical organization can be reproduced without assuming a specific plaintext, and what kind of materially simple procedure could implement the relevant constraints without reliance on modern computational technology?

This change of emphasis arose from exploratory work on the manuscript’s word-form structure. Simple substitution models did not appear to provide a natural explanation for several recurring properties: restricted character adjacency, highly structured word forms, strong low-order transition patterns, paragraph-initial galls, line- and paragraph-dependent effects, and persistent physical-bifolium signatures. These observations suggested that it would be useful to separate two questions that are often conflated: whether the VM ultimately encodes linguistic meaning, and

whether its visible character strings could have been produced through a constrained combinatorial procedure.

The reconstruction developed below addresses only the second question. It is therefore best understood as a constructive historical model. If a materially simple procedure can reproduce a substantial subset of the manuscript’s statistical properties, this does not prove that the procedure was historically used. It does, however, establish that those properties do not by themselves require modern computation, nor do they necessarily require the direct storage of complete Voynich words.

The remainder of the paper is organized as follows. Section 2 presents the methodological development of the reconstruction. It first defines the dataset and the external Currier A/B and scribal-hand classifications, then describes the exploratory path from the Lullian combinatorial analogy to the twenty-three-character seven-layer representation. Further analysis shows how this layered representation can be understood as a coarse-grained manifestation of lower-order n -gram structure and how this, in turn, leads to the two-character state model that inspired the physical shelf hypothesis. Section 3 presents the shelf-and-tablet apparatus, its treatment of SPACE and paragraph starts, and the computational optimization and fit assessment of the A-2 reconstruction. Section 4 discusses the broader implications of the reconstruction, including the unused state space and the possibility of apparatus reuse across scribes or Currier regimes. Section 5 summarizes the constructive result and identifies the main questions for further testing.

2 Methodological Background and Development of the Reconstruction

2.1 Dataset and external classifications.

The textual analyses use the ZL transliteration file (ZL3b), Version 3b of 13 May 2025, as their principal dataset [2, 3]. ZL3b is treated as a modern transliteration rather than as the manuscript itself. The results are therefore conditional on its character segmentation, word-boundary decisions, treatment of uncertain readings, and other transliteration conventions. Such dependence is particularly important in the Voynich Manuscript, for which the identification and segmentation of individual characters remain partly interpretive [3].

For the principal analyses and the physical reconstruction, the working inventory was restricted to the twenty-three VM character types occurring at least 100 times in the working whole-manuscript corpus:

$$\{q, cfh, ch, cph, cth, p, sh, ckh, f, k, o, t, a, d, e, s, i, l, y, r, m, n, g\}.$$

The sequences *cfh*, *ch*, *cph*, *cth*, *sh*, and *ckh* are each treated as single VM character units rather than as sequences of separate characters. The frequency threshold is used here only to define a stable working inventory for statistical comparison and mechanical reconstruction; it is not intended as a paleographic claim that these twenty-three units constitute the manuscript’s complete historical character inventory. Rarer

signs and uncertain or alternative readings are excluded from analyses restricted to this working inventory, and uncertain or alternative readings are also excluded in the stricter statistical subsets where appropriate.

Two established external classifications are used. Currier A and B follow Prescott Currier’s statistical distinction between two bodies of Voynich text [4]. Currier’s term “language” is treated here only as a conventional label and not as evidence that two underlying natural languages have been identified; this neutral usage is also consistent with later reviews [1].

Scribal-hand assignments follow the five-hand paleographic classification proposed by Lisa Fagin Davis [5]. These assignments concern the physical hands responsible for writing the manuscript and are not assumed to identify authorship. Their statistical detectability has subsequently received computational support from Farrugia, Layfield, and van der Plas, who showed that character bigrams and trigrams can predict several of the Davis hand labels substantially above chance [6]. Currier A/B and scribal hand are therefore treated as separate external covariates throughout this study.

2.2 Computational procedures.

All computational analyses were performed on VM-character sequences obtained from the ZL3b transcription under the segmentation conventions specified above. Unless otherwise stated, character n -grams were counted only within individual transcribed words and did not cross word, line, or paragraph boundaries. The units *cfh*, *ch*, *cph*, *cth*, *sh*, and *ckh* were each treated as single characters.

Several different evaluation procedures are used below, and they should be distinguished.

For the twenty-three-character seven-layer bifolium classification, the analysis was restricted to 21 complete Herbal A bifolia belonging to Currier A and Hand 1. Each bifolium contains four surfaces. Four fixed hold-out folds were used: in each fold, the same surface position was withheld from all 21 bifolia, leaving 63 surfaces for feature weighting and 21 withheld surfaces for evaluation. For each candidate bifolium, the remaining three surfaces were combined to form its prototype. Inverse-document-frequency weights were estimated from the 63 training surfaces only, and withheld surfaces were ranked against the 21 bifolium prototypes by cosine similarity. Top-1 and Top-3 accuracy were calculated from these rankings. For the one-to-one measure, the 21 withheld surfaces were jointly assigned to the 21 candidate bifolia by maximum-weight bipartite assignment (Hungarian algorithm), so that no two withheld surfaces could be assigned to the same bifolium.

The comparison of word-internal bigrams, trigrams, 4-grams, and 5-grams used a separate repeated hold-out procedure. The same 2,000 hold-out configurations were used for all four n -gram orders so that differences between them were not caused by different selections of withheld surfaces. In each configuration, one of the four surfaces of every bifolium was withheld and the remaining three were used to construct the corresponding bifolium prototype. n -gram counts were represented with sublinear TF-IDF weighting, normalized at the surface level, and the three training surfaces were averaged to form each prototype. Classification was again based on similarity between the withheld surface and the candidate bifolium prototypes. The reported

31.27% bigram Top-1 accuracy, for example, is the mean over these repeated controlled hold-out configurations rather than the result of a single four-fold partition.

The next-state prediction experiment used page-grouped five-fold cross-validation on Herbal A. Pages, rather than individual words or character sequences, were assigned to folds, so that all observations from a held-out page remained outside the corresponding training set. The predictive sequences were augmented with computational START and END boundary states, giving 38,547 next-state predictions across the 95 Herbal A pages and 7,694 analyzed words. The bigram model conditioned on the immediately preceding state, whereas the trigram model conditioned on the preceding two-state context. Predictive performance was measured as mean negative log-likelihood in bits per next state.

The START and END symbols used in this predictive experiment are computational boundary markers only. They are not included in the word-internal n -gram counts used for the bifolium-classification analysis, and they should not be confused with the SPACE state or the separate Paragraph START box introduced later in the physical reconstruction.

Randomization tests are described separately where they are used. In particular, the seven-layer directionality comparison preserves the observed layer-size profile while randomly reassigning the 23 characters among those layers. Because the seven-layer ordering itself was developed through exploratory optimization on related Voynich material, the resulting randomization value is treated as a descriptive benchmark for the extremity of the final ordering rather than as a fully independent confirmatory p -value.

2.3 Seven-Layer Representation as an Exploratory Precursor

Academic articles generally omit much of the personal and exploratory process by which an author arrives at a particular hypothesis. In the present study, however, the reconstruction proposed below emerged from a line of inquiry somewhat different from the approaches more commonly applied to the Voynich Manuscript. The author therefore considers it useful to explain, in condensed form, why this particular line of investigation was pursued. The following account is not intended as evidence for the reconstruction itself, but rather as a description of the methodological path that led to it.

The author's field is the history of ancient and early modern European law. In addition to working with English, German, French, Latin, and Greek, the author is also familiar, as a Japanese scholar, with the textual and graphic traditions of the Chinese-character cultural sphere. At an early stage of the investigation, he therefore considered whether the VM might be understood as a relatively straightforward substitution or comparable transformation of one of these familiar linguistic systems. A preliminary examination of its word-form structure and patterns of repetition, however, gave the impression that at least its principal statistical characteristics would be difficult to account for naturally by a simple substitution model.

Several exploratory tests were also conducted, with the assistance of ChatGPT 5.6, on Semitic languages outside the author's own principal expertise, particularly Arabic and Hebrew. Following these preliminary investigations, the author decided

not to make the search for a specific natural-language plaintext and its corresponding substitution rules the principal line of inquiry in the present study. This is not intended to claim that the VM cannot be interpreted or translated as a natural language. Natural-language decipherment hypotheses remain outside the scope of the present reconstruction.

Through work on early modern European intellectual and legal history, the author had become familiar with Gottfried Wilhelm Leibniz (1646–1716) and, indirectly through that context, with the Lullian art. The Art of Ramon Llull (c. 1232–1315/16) employed a finite inventory of symbolic elements together with systematic procedures for combining them. Among its diagrammatic devices were revolving figures or volvelles; in the later form of the Art, the Fourth Figure used such a revolving arrangement to generate ternary combinations [7, pp. 22–23]. Leibniz is relevant here only as the author’s route of acquaintance with this tradition; the historically relevant precedent for the present exploratory analogy is the medieval Lullian combinatorial tradition itself.

A simple schematic example illustrates the combinatorial principle. If two lettered terms are denoted A and B , their relation can be examined in both directions, AB and BA . In Llull’s own example, using the letters B and C , the corresponding propositions include “goodness is great” and, in the reverse direction, “greatness is good” [7, p. 147]. The notation AB/BA is used here only as a simplified illustration; Llull’s Third Figure represents the underlying binary combination as a single compartment and derives the directional propositions from it.

During the exploratory phase described above, before the shelf model was formulated, the author, with the assistance of ChatGPT and through iterative optimization, attempted to organize the recurrent VM signs into a combinatorial map analogous to Llull’s rotating figures, in order to make common character paths easier to inspect by eye.

After a number of exploratory revisions and unsuccessful arrangements, this process eventually yielded the following particularly interesting seven-layer ordering for a twenty-three-character working inventory:

$$\begin{aligned} L0 &= \{q\}, \\ L1 &= \{cfh, ch, cph, cth, p, sh\}, \\ L2 &= \{ckh, f, k, o, t\}, \\ L3 &= \{a, d, e, s\}, \\ L4 &= \{i, l, y\}, \\ L5 &= \{r\}, \\ L6 &= \{m, n, g\}. \end{aligned}$$

In the controlled sample of 21 complete Herbal A bifolia belonging to Currier A and Hand 1, comprising 84 surfaces, the twenty-three-character seven-layer representation retained the same strongly directional structure. Of all word-internal transitions within the covered vocabulary, 61.64% moved forward through the ordering, 23.23% remained within the same layer, and only 15.13% moved backward.

To evaluate how unusual this directionality is relative to arbitrary arrangements having the same layer sizes, the twenty-three characters were randomly reassigned 50,000 times while preserving the layer-size profile 1, 6, 5, 4, 3, 1, 3. The mean backward-transition rate under these randomized arrangements was 39.65%. Only one of the 50,000 randomized arrangements produced a backward-transition rate as low as or lower than the observed value. With the usual plus-one correction, the corresponding randomization tail frequency is approximately

$$\frac{1 + 1}{50,000 + 1} \simeq 4.0 \times 10^{-5}.$$

Because the seven-layer ordering was itself developed through exploratory optimization on related Voynich material, this value is reported as a descriptive randomization benchmark for the extremity of the final ordering, rather than as an independent confirmatory *p*-value.

The effect was also consistent at the individual-bifolium level. Across the 21 controlled bifolia, observed backward-transition rates ranged from 10.39% to 19.19%, and all 21 were substantially below their corresponding randomized expectations.

Table 1 summarizes the results for this controlled sample.

Table 1 Twenty-three-character seven-layer representation in the controlled Herbal A, Currier A, Hand-1 sample

Measure	Result
Complete bifolia	21
Surfaces	84
Analyzed word tokens	6,765
Covered word tokens	6,747
Token coverage	99.73%
Observed word types	2,012
Covered word types	1,994
Type coverage	99.11%
Word-internal transitions	20,207
Forward transitions	61.64%
Within-layer transitions	23.23%
Backward transitions	15.13%
Randomized backward transitions	39.65%
Words with no backward transition	61.72%
Words with zero or one backward transition	93.42%
Backward range across bifolia	10.39–19.19%
Descriptive randomization tail frequency	4.0×10^{-5}

It then emerged that the same seven-layer representation could be used to classify physical bifolia substantially above chance. To test this, the analysis was restricted to the 21 complete Herbal A bifolia belonging to Currier A and Hand 1, giving 84 surfaces in total. The controlled sample contained 6,765 word tokens, of which 6,747 were covered by the twenty-three-character seven-layer representation.

For each of the four surface positions in turn, that surface was withheld from every bifolium. The remaining three surfaces of each bifolium were used to construct its

prototype, and the 21 withheld surfaces were then classified among the 21 candidate bifolia. Feature weights were estimated from the 63 training surfaces only.

Table 2 gives the resulting classification performance.

Table 2 Bifolium classification using the twenty-three-character seven-layer representation

Model	Top 1	Top 3	One-to-one
Layer structure only	27.38%	57.14%	22.62%
Full seven-layer representation	38.10%	76.19%	42.86%
Chance level	4.76%	14.29%	4.76%

The layer-structure model uses starting and ending layers, layer usage, layer-to-layer transitions, within-layer continuation, backward movement, and related coarse seven-layer properties, without using complete words or character n -grams. Even under this restriction, the correct bifolium was the first-ranked candidate in 27.38% of cases, compared with a chance level of 4.76%.

The full seven-layer representation additionally retains which character is selected within each layer and which character-to-character port is used when moving between layers. With this information, Top-1 accuracy increased to 38.10%, Top-3 accuracy to 76.19%, and one-to-one assignment accuracy to 42.86%.

Thus the seven-layer representation retains substantial information about physical bifolium membership even after Currier class and scribal hand have been held constant. The improvement obtained by restoring within-layer character choices and inter-layer ports further indicates that bifolium-specific variation is carried not only by coarse layer usage but also by the local weighting of pathways within the shared seven-layer structure.

The hold-out procedure prevents the withheld surfaces themselves from contributing to feature weighting or to the construction of their bifolium prototypes. It does not, however, constitute a fully independent validation of the seven-layer representation itself. The assignment and ordering of characters among the seven layers were developed exploratorily on related Voynich material before this classification experiment. Accordingly, Table 2 should be interpreted as controlled descriptive evidence that the resulting representation retains bifolium-specific information, rather than as a fully independent test of a representation fixed in advance. A stricter external validation would require fixing the seven-layer representation beforehand and applying it unchanged to additional bifolia not involved in its development.

2.4 From the Seven-Layer Heuristic to the Shelf Hypothesis

Further investigation of the seven-layer representation showed that it was not an independent structure imposed on the manuscript. Rather, the ordered layers appeared to be a coarse-grained manifestation of the same local transition regularities that are visible more directly in the character n -gram distributions. In this sense, the seven-layer map can be understood as a higher-level structure emerging from, or “bleeding through” from, the underlying n -gram organization.

This realization shifted the analysis from the seven-layer map itself to the short-range transition system that appeared to generate it. The next step was therefore to examine the character n -grams directly, especially the local two- and three-character dependencies and their variation across physical bifolia.

First, short-context models showed that much of the observable structure can be represented by local conditional choices. At trigram level, the next character can be modeled from the preceding two-character state,

$$[a, b] \longrightarrow c.$$

The predictive advantage of retaining this second preceding character is measurable on unseen Herbal A pages. In page-grouped five-fold cross-validation, with computational START and END states included so that word boundaries were scored as part of the next-state prediction, a bigram model conditioning on the immediately preceding state gave an average log-loss of approximately

$$2.2931 \text{ bits/next state,}$$

whereas the trigram model conditioning on the preceding two-state context reduced this to

$$2.1691 \text{ bits/next state.}$$

Thus retaining the additional contextual state improved out-of-sample prediction by approximately 0.124 bits per next state. The relevant local information is therefore not exhausted by a simple character-to-character transition: the state $[a, b]$ contains additional predictive information about the following character c .

For the mechanical interpretation below, attention is then restricted to the ordinary character states themselves; START and END served only as computational boundary markers in the predictive comparison.

For each observed two-character context $[a, b]$, the transcription therefore supplies an empirical distribution over possible next characters,

$$P(c \mid a, b) = \frac{N(a, b, c)}{\sum_x N(a, b, x)}.$$

This statistical representation has an immediate mechanical interpretation. A physical location can represent the state $[a, b]$, and movable tablets stored at that location can represent the possible outputs. Their multiplicities provide an integer approximation to $P(c \mid a, b)$. Drawing a tablet marked c writes the next character and updates the state:

$$[a, b] \longrightarrow c \longrightarrow [b, c].$$

The same operation can then be repeated from the new state. A local three-character dependency can therefore be implemented without storing complete Voynich words and without requiring a three-dimensional table: the apparatus needs only a set of two-character states and locally weighted choices of the next character.

Second, the physical bifolium emerged as an important unit of variation in these same low-order transition statistics. This was tested by withholding one surface from

each complete bifolium and attempting to identify its bifolium from the remaining three surfaces. To remove the major confounds of Currier class, scribal hand, and broad subject matter, the principal test was restricted to 21 complete Herbal A bifolia belonging to Currier A and Hand 1.

The classification performance was far above the $1/21 = 4.76\%$ chance level. Using word-internal bigrams alone, the withheld surface was assigned to the correct bifolium as the first-ranked candidate in

$$\boxed{31.27\%}$$

of trials. Under a one-to-one assignment in which each withheld surface had to be matched to a different bifolium, accuracy was

$$\boxed{29.11\%},$$

and the correct bifolium appeared among the three highest-ranked candidates in

$$\boxed{61.77\%}$$

of trials. The corresponding Top-1 accuracy for trigrams was 25.59%, and performance declined further for 4-grams and 5-grams.

The fact that bigrams outperform longer n -grams is important for the interpretation. It suggests that the bifolium-specific signal is concentrated primarily in the relative weighting of very local character transitions rather than in long characteristic strings or memorized complete words.

The first and second results therefore describe two complementary properties of the same system. The trigram result shows that the next character is better predicted when the current two-character context is retained,

$$[a, b] \longrightarrow c,$$

whereas the bifolium-classification result shows that the frequencies with which these local transitions are used vary systematically between physical bifolia.

The bigram classification result does not by itself establish that the conditional distributions $P(c \mid a, b)$ differ between bifolia. Strictly speaking, it establishes that low-order transition frequencies carry bifolium-specific information. The reconstruction below therefore tests the stronger constructive hypothesis that such bifolium-level variation can be represented by changing tablet multiplicities within a common two-character state space.

Their intersection suggested the material decomposition used below:

$$\boxed{\text{fixed two-character state space} \quad \times \quad \text{bifolium-specific transition weights}}$$

A compartment at coordinate $[a, b]$ can represent the common local state. The possible next characters can be represented by movable tablets stored in that compartment, while their multiplicities can be changed from one bifolium to another. The shelf itself therefore remains fixed, whereas its local output weights are reconfigured.

Schematically,

shelf geometry $\longrightarrow$ shared two-character state space,
 tablet multiplicities $\longrightarrow$ bifolium-specific transition weighting.

This conjunction—a common short-context state system together with bifolium-specific weighting of its transitions—was the decisive conceptual step from the statistical analysis to the physical shelf hypothesis.

3 The Shelf-and-Tablet Apparatus

3.1 Preparation of the Main Shelf

The main apparatus is a wooden open-compartment shelf divided into a 24×24 grid, giving 576 compartments. The same twenty-four states are written on the horizontal and vertical axes: SPACE plus the twenty-three principal VM character types.

Each compartment $[a,b]$ represents the two immediately preceding states. Thus $[q,o]$ represents the state in which the active context is q followed by o , while $[y,SPACE]$ represents a word ending in y followed by a space. The physical position of the compartment therefore stores the context; the tablets stored inside need not repeat it.

Each movable tablet carries only (1) one VM character and (2) one stopping number. In the present implementation, a tablet may be written schematically as $c[n]$. Multiple identical tablets may occupy the same compartment. Their multiplicities implement relative transition weights physically. Once the shelf is prepared, the operator need not calculate a probability.

There are no separate END tablets or END compartments. After a character has been written, the tablet's number n is compared with the current length k of the word. If $n \leq k$, the word ends and a space is written. If $n > k$, generation continues.

The twenty-four shelf states are ordered broadly with reference to the twenty-three-character seven-layer representation described in Section 2, with SPACE added as an additional mechanical boundary state. This ordering is convenient and reduces large physical reversals in many ordinary word-internal paths, but the shelf mechanism itself does not require the operator to reason explicitly in terms of seven layers.

3.2 SPACE and Ordinary Word-Boundary Operation

SPACE is treated as a genuine mechanical state.

Suppose a word ends in character c . The apparatus enters $[c,SPACE]$. The identity of the word-final character is therefore retained across an ordinary word boundary. From $[c,SPACE]$, the first character d of the next word is selected. The new state becomes $[SPACE,d]$. After the second character e is selected, the apparatus enters $[d,e]$ and resumes ordinary two-character operation.

In schematic form:

$[a,b] \rightarrow c \rightarrow [b,c] \rightarrow d \rightarrow [c,d] \rightarrow \dots$

and, at a word boundary:

$[a,b] \rightarrow c[n] \rightarrow [c,SPACE] \rightarrow d \rightarrow [SPACE,d] \rightarrow e \rightarrow [d,e]$.

This permits a relation such as -y followed by q- to be implemented directly by increasing the proportion of q tablets in [y,SPACE]. No separate cross-word rule is needed.

The special compartment [SPACE,SPACE] is a neutral restart state. It is used when immediate word-final-to-word-initial memory is to be suppressed, as in the line-related restart positions modeled in A-2. It is not the paragraph-start mechanism.

3.3 Paragraph START

Paragraph beginnings require a separate mechanism because a neutral restart alone does not reproduce the pronounced concentration of gallows characters at paragraph starts.

The final reconstruction therefore uses a single Paragraph START box, employed only once at the beginning of each paragraph. Its inventory is weighted toward the paragraph-initial gallows distribution of the relevant bifolium. After the first character has been selected, the operator moves immediately to the corresponding [SPACE,character] compartment of the main shelf. The Paragraph START box is not used again until the next paragraph begins.

The roles are therefore deliberately separated:

ordinary word boundary $\rightarrow$ [c,SPACE] neutral line-related restart $\rightarrow$ [SPACE,SPACE] paragraph beginning $\rightarrow$ Paragraph START.

This separation is essential. If the paragraph START distribution were reused after every line break, the model would predict excessive gallows at ordinary line beginnings.

3.4 Operating Procedure and Optional Weak Temporal Bias

The depth of a compartment also permits a possible weak non-stationary refinement. Used tablets could, for example, be returned preferentially toward the front, so that subsequent draws become slightly biased toward recently used paths. Exploratory simulations indicate that a weak front bias introduced after the first line, together with returning used tablets toward the front at line end, can produce the same qualitative increase in related-word carryover from the second to the third line that is observed in Herbal A. Because this temporal refinement was explored separately and is not required for the static A-2 fit assessed below, it is treated here only as a candidate secondary mechanism rather than as a validated component of the minimal reconstruction.

3.5 Computational Optimization and A-2 Fit Assessment

3.5.1 A-2 Configuration

A-2, consisting of f2r, f2v, f7r, and f7v, was used as a concrete bifolium-level test case. The aim was not to infer the historical apparatus directly from these four surfaces, but to ask whether the observed A-2 statistics could be reproduced by a compact instance of the general shelf architecture.

A-2 contains eight paragraph beginnings. All eight are gallows-initial: *k* occurs six times, *f* once, and *p* once. The Paragraph START box for this reconstruction therefore contains eight tablets, *k*[8] × 6, *f*[8] × 1, and *p*[8] × 1. This is a within-bifolium reconstruction of the observed distribution, not an independent prediction.

The A-2 apparatus contains 576 main-shelf compartments plus one Paragraph START box, for 577 physical compartments. Only a minority contain tablets. The selected reconstruction, whose inventory size is justified below, uses 600 main-shelf tablets plus eight START tablets, for 608 tablets in total.

The 600 main-shelf tablets consist of 294 internal-state tablets, 146 tablets in [SPACE, *x*] second-character states, 111 tablets in [*x*, SPACE] ordinary word-boundary states, 35 tablets in [SPACE, SPACE], and 14 minimal safety tablets placed in otherwise unobserved boundary states. The fourteen safety tablets prevent the finite simulation from entering an unrecoverable state and are explicitly treated as a technical completion, not as an empirical historical inference.

3.5.2 Optimization procedure.

The A-2 configuration was fitted to the 297 clean word tokens on *f2r*, *f2v*, *f7r*, and *f7v*. Complete-word identities were not used as entries in the apparatus or as an accept–reject criterion during generation. Instead, the fitting procedure operated on the local events required by the shelf mechanism.

The observed events were divided into four mechanically distinct classes. INTERNAL events correspond to ordinary two-character states [*a*, *b*] followed by a next character *c*. SECOND events correspond to the selection of the second character from a state [SPACE, *b*]. BOUNDARY events correspond to the selection of the first character of the next word from [*b*, SPACE]. Finally, [SPACE, SPACE] was treated separately as the neutral restart state. Within each class, the observed A-2 frequencies supplied the relative weights of the available next-character choices, and these continuous weights were converted into integer tablet multiplicities subject to a fixed total inventory. Stopping numbers were fitted simultaneously from the observed distribution of stopping and continuation at different word depths, under the single rule that a tablet *c*[*n*] terminates the current word when $n \leq k$, where *k* is the word length after *c* has been written.

Paragraph beginnings were not included in this optimization. Their eight first-character observations were represented separately and exactly by the Paragraph START inventory *k*[8] × 6, *f*[8] × 1, and *p*[8] × 1. Likewise, the neutral restart inventory was fitted separately to the observed A-2 restart distribution. Fourteen otherwise unobserved [*b*, SPACE] states were then given a single safety tablet each. These safety tablets are a technical completion introduced only to prevent an unrecoverable state in finite simulations and were not counted as empirical evidence for an A-2 transition.

In the simulation-based A-2 assessment, “A-2-sized” means that each run produced 297 word tokens. The simulations did not attempt to generate the higher-level page, paragraph, or line layout *de novo*. Paragraph beginnings and line-related neutral restarts were supplied according to the same A-2 boundary procedure used in fitting. Paragraph START was invoked only at paragraph beginnings, whereas [SPACE, SPACE] was used at neutral line-related restart positions. Ordinary word

boundaries continued to pass through $[c, \text{SPACE}]$, and therefore retained the preceding word-final character.

The fit statistics reported below should consequently be interpreted as tests of stochastic word-form, word-length, and local transition behavior conditional on this supplied A-2 boundary structure. They are not independent predictions of the number or placement of paragraph and line boundaries. The weak front-bias refinement described in Section 3.4 was not used in these static A-2 fit assessments.

To determine the appropriate integer resolution, the same static shelf architecture was reconstructed with main-shelf budgets of 475, 500, 525, 550, 575, 600, 625, and 650 tablets, while the eight Paragraph START tablets were held fixed. For each budget, A-2-sized samples were generated and compared with the observed material using several diagnostics rather than a single fitted statistic: word-type and hapax counts, mean word length and its dispersion, the frequency of very short and long words, and Jensen–Shannon divergence for the bigram, trigram, and word-boundary distributions.

Increasing the main-shelf inventory from 475 to 600 tablets produced substantial and fairly consistent improvements in both lexical diversity and low-order transition fit. Beyond 600 tablets, however, the improvement was small. In particular, increasing the inventory to 625 or 650 tablets produced essentially no further reduction in trigram divergence and did not remove the persistent hapax deficit. The value 600 was therefore selected as a practical elbow of the inventory–fit curve, not as a uniquely optimal or historically privileged number.

For the selected configuration, the 600 main-shelf tablets were distributed as 294 INTERNAL tablets, 146 SECOND tablets, 111 BOUNDARY tablets, 35 tablets in $[\text{SPACE}, \text{SPACE}]$, and 14 safety tablets. Thus the optimization determines an integer approximation to the observed local transition and stopping structure under a fixed material budget; it does not optimize against a stored list of complete Voynich words. Because the same A-2 material was used to construct these multiplicities, the resulting simulations constitute an in-sample fit assessment rather than an external predictive validation.

3.5.3 Simulation-Based Fit Assessment and Attested-Word Recovery

The fitted 608-tablet configuration was first assessed by means of 1,000 independent A-2-sized simulation runs.

The principal results of this simulation-based fit assessment are summarized in Table 3. The observed A-2 sample contains 297 word tokens, 195 word types, and 155 hapax legomena. Across the 1,000 simulated A-2-sized samples, the corresponding mean values were 188.54 word types and 136.17 hapax legomena. Mean word length was 4.0186 characters in the simulations, compared with 3.9865 in A-2, while the standard deviation of word length was 1.6186 compared with 1.5763.

The fitted apparatus therefore reproduces the central tendency and dispersion of A-2 word length closely, while also retaining relatively small divergences in its low-order character distributions. The main remaining discrepancy is lexical diversity: the simulated samples produce fewer hapax legomena than observed in A-2. This deficit persists as the tablet inventory is increased and is therefore treated below as evidence

Table 3 Simulation-based fit assessment of the 608-tablet A-2 reconstruction

Measure	A-2 observed	Simulation mean
Word types	195	188.54
Hapax legomena	155	136.17
Mean word length	3.9865	4.0186
Word-length SD	1.5763	1.6186
One-character words	4.71%	4.46%
Words of eight or more characters	1.35%	1.17%
Bigram JSD	—	0.03892
Trigram JSD	—	0.10608
Word-boundary JSD	—	0.11462
Paragraph-start gallows	100.00%	100.00%
Neutral-restart gallows	2.86%	2.77%

of residual model structure rather than merely insufficient integer resolution. The paragraph-start gallows result is a fitted property of the dedicated Paragraph START inventory and is not an independent prediction.

A further 5,000 independent simulation runs were used to measure the rate at which the fitted apparatus reproduced complete word forms attested in the manuscript. This is a simulation-based assessment of the fitted A-2 reconstruction rather than an external validation, because A-2 itself was used in optimizing the tablet configuration. The comparison with forms attested outside A-2 nevertheless provides a separate measure of how often the local mechanism generates word forms not present in its fitting material but attested elsewhere in the manuscript.

Across the 5,000 runs, 55.00% of generated tokens matched forms attested in A-2 itself. An additional 17.52% matched forms absent from A-2 but attested elsewhere in Herbal A, and a further 4.53% matched forms absent from Herbal A but attested elsewhere in the strict whole-manuscript comparison vocabulary. The remaining 22.95% were unattested in that comparison set. Thus,

$$55.00\% + 17.52\% + 4.53\% = 77.05\%$$

of generated tokens matched a complete form attested somewhere in the manuscript, of which 22.05 percentage points represented forms attested outside the A-2 fitting material.

These figures are properties of the optimized simulation and are not independent estimates of out-of-sample predictive accuracy. In particular, the 55.00% A-2 component should be interpreted as within-sample recurrence. The 22.05% attested outside A-2 is the more relevant quantity when asking whether the local mechanism generates Voynich forms beyond those directly used to fit the reconstruction. No complete-word dictionary or rejection filter was used in generation.

3.6 Comparison with matched low-order null generators.

The high complete-word attestation rate produced by the shelf does not, by itself, establish that complete-word recurrence is specific to the physical reconstruction. To determine how much of this effect follows from low-order character structure

alone, three matched stochastic null models were fitted to the same A-2 material and evaluated against the same attestation vocabularies.

The corpus extraction used for this comparison reproduced the datasets used above exactly: 297 A-2 tokens representing 195 word types, 7,694 Herbal A tokens representing 2,270 word types, and a strict whole-manuscript comparison vocabulary of 37,597 tokens and 7,455 word types.

For the primary comparison, each null model generated 297 words in each of 5,000 independent runs. The observed multiset of A-2 word lengths was supplied directly to every null generator. The comparison is therefore deliberately conservative with respect to the shelf: the null models were not required to reproduce the A-2 word-length distribution, whereas word termination in the shelf reconstruction remained a consequence of its numbered-tablet stopping mechanism.

The unigram null sampled characters from the A-2 character-frequency distribution. The bigram null used first-order character-transition probabilities, and the trigram null used the preceding two-character context. For the bigram and trigram models, START and END probabilities were retained and generation was conditioned on the supplied target word length. None of the null models stored or consulted a complete-word lexicon.

Table 4 compares the resulting complete-word attestation rates with those of the 608-tablet shelf reconstruction.

Table 4 Complete-word attestation under the A-2 shelf reconstruction and length-matched low-order null generators. Each null result is the mean of 5,000 independent 297-token simulations.

Model	A-2	Herbal A only	Other VM	VM total	Outside A-2	Unattested
608-tablet shelf	55.00%	17.52%	4.53%	77.05%	22.05%	22.95%
Unigram null	3.40%	5.99%	2.98%	12.37%	8.97%	87.63%
Bigram null	48.66%	23.71%	4.89%	77.26%	28.60%	22.74%
Trigram null	64.82%	19.03%	3.77%	87.62%	22.80%	12.38%

The result changes the interpretation of the 77.05% shelf-attestation rate. A unigram model reproduces only 12.37% attested VM tokens, showing that simple character frequencies are insufficient. Once local transition structure is retained, however, complete-word recurrence rises sharply. The length-matched bigram null reaches 77.26%, essentially the same overall VM-attestation rate as the shelf, while the trigram null reaches 87.62%.

The same point is visible when forms already attested in A-2 are removed. The shelf generates 22.05% tokens that are absent from A-2 but attested elsewhere in the manuscript. The corresponding values are 28.60% for the bigram null and 22.80% for the trigram null. Thus recurrence of complete forms outside the fitting material is not, by itself, a shelf-specific effect. A substantial amount of such recurrence follows naturally from the manuscript’s low-order character constraints.

This result is important for the interpretation of the reconstruction. The shelf should not be understood as outperforming a statistical trigram model in recovering

known Voynich word forms. Rather, it provides a finite material implementation of much of the same local structure without storing complete words.

As a sensitivity analysis, the three null models were also allowed to generate their own word lengths from their fitted START/END behavior, rather than being supplied with the observed A-2 lengths. The qualitative ordering remained unchanged. Overall VM-attestation rates were 32.01% for the unigram model, 75.77% for the bigram model, 86.69% for the trigram model, and 77.05% for the shelf reconstruction.

The free-length comparison nevertheless reveals an important residual advantage of the numbered-tablet mechanism. Mean word length alone was reproduced by both low-order Markov models, but their word-length distributions were substantially broader than that of A-2. The observed word-length standard deviation is 1.5763, compared with 1.6186 for the shelf, approximately 1.81 for the free-length trigram model, and approximately 2.52 for the free-length bigram model. Similarly, words of eight or more characters constitute 1.35% of observed A-2 tokens and 1.17% of shelf output, but approximately 4.40% under the trigram null and 9.05% under the bigram null.

Lexical diversity shows a related difference. A-2 contains 195 surface types in 297 tokens. The shelf produces a mean of 188.54 types, whereas the free-length trigram null produces approximately 168 types and the free-length bigram null approximately 174. The trigram model therefore achieves a higher complete-word attestation rate partly by concentrating probability on a smaller set of high-probability forms.

Taken together, these comparisons suggest a more specific role for the physical reconstruction. High complete-word recurrence is largely a consequence of the VM's low-order transition structure and should not be treated as independent evidence for the shelf mechanism. The constructive result is instead that this low-order structure can be discretized into a simple material state machine while simultaneously maintaining a substantially narrower VM-like word-length distribution and greater lexical diversity than the corresponding unconstrained low-order Markov generators.

3.7 Cross-Currier Reuse of Active Compartments in Mixed Herbal Quires

The shelf interpretation also permits a direct test of whether material belonging to different Currier regimes tends to reuse the same local two-character state space within a physical quire.

The test was restricted to the mixed Herbal quires D–G. The A bifolia in these four quires were held fixed, while the six B bifolia were reassigned among D–G subject to the observed numbers of B bifolia per quire,

$$1 \mid 2 \mid 2 \mid 1.$$

All

$$\frac{6!}{1! 2! 2! 1!} = 180$$

possible reallocations satisfying this constraint were evaluated.

For this analysis, a character-pair compartment was counted as *active* when the corresponding two-character state was followed by at least one observed next-character

transition in the relevant material. For each allocation, the number of active compartments required by the B material but not already active in the A material of the same quire was then counted. Thus the statistic measures the amount of additional shelf state space that would have to be activated when moving from the A material to the B material within each quire.

In the observed arrangement, the six B bifolia required a total of

99

new active compartments beyond those already used by the A material of their respective quires. Across the 180 admissible reallocations, the mean was

108.17.

The observed arrangement lay at the low end of the exact reassignment distribution, giving the one-sided reassignment probability

$p = 0.0056$.

Thus the actual placement of the B bifolia reuses more of the active two-character compartment structure of the A material in the same quire than would typically occur under reassignment of the same six B bifolia.

This result concerns the *availability of local states*, not the weights of the tablets stored within those states. It therefore does not show that A and B used identical transition probabilities, nor does it establish that one scribe physically inherited an apparatus directly from another. More conservatively, it shows that the observed A/B arrangement is compatible with reuse of a substantial part of a common shelf geometry while allowing the local tablet multiplicities to be reconfigured.

4 Discussion

The reconstruction supports several limited conclusions.

First, a substantial portion of Voynich surface structure can be reproduced from low-order local constraints. This includes word-length statistics, low-order character-transition distributions, and a substantial amount of complete-word recurrence.

Second, the economy of the reconstruction suggests a change in research priority. If a comparatively simple apparatus—a 24×24 open-compartment shelf, a single dedicated Paragraph START box, and only about six hundred movable tablets—can already reproduce this much of the observed surface structure, then direct cryptanalytic decipherment should not be treated as the only urgent next step. An equally urgent task is to determine whether a progressively more constrained mechanism can reproduce the manuscript as a whole with still higher precision. Such a program would provide explicit null models for decipherment claims and could reveal structural regularities that are difficult to see when the manuscript is approached only as a ciphertext.

The present reconstruction suggests at least two concrete directions for that program.

- **The unused state space may itself contain information.** The shelf has many more geometrically possible compartments than are actually used. In the current 23-character pair inventory, 207 of the $23 \times 23 = 529$ character-pair states are unattested before SPACE states are added. Unattested does not by itself mean structurally forbidden: some absences may reflect limited observation or transliteration choices. A priority for future work is therefore to test whether some or most of this sparsity has a compact mathematical description—for example, a graph, layered order, product structure, or other constraint inherited from the generating mechanism—rather than treating every unobserved pair as either an independent accident or a proven prohibition.
- **The apparatus may have been reusable across scribes or Currier regimes.** Codicological reconstructions have previously argued that the present order of some Voynich bifolia need not preserve their original arrangement, and the apparently mixed Herbal A and Herbal B bifolia have been discussed in this connection [8]. The exact reassignment test reported in Section 3.7 shows that, within the mixed Herbal quires D–G, the B material in its observed locations reuses more of the A material’s active compartment structure than expected under admissible reallocations of the same B bifolia. This does not establish chronological transfer of a physical apparatus, but it is consistent with a model in which much of a common shelf geometry was retained while local tablet weights were reconfigured across Currier regimes.

5 Conclusion

The central result of this study is constructive. The VM’s low-order character structure, bifolium-specific weighting, word-boundary dependencies, and much of its word-form distribution can be approximated by a compact physical state machine implemented as an open-compartment shelf with movable numbered character tablets. In the A-2 implementation, the paragraph-initial gallows distribution is represented separately by a dedicated Paragraph START inventory fitted directly to the eight observed paragraph beginnings.

The minimal logic is simple. Shelf coordinates store two-state context. Tablet multiplicities encode weighted next-character choices. A number printed on each tablet supplies a stopping threshold. SPACE preserves ordinary word-boundary state. [SPACE,SPACE] supplies a neutral restart. A separate Paragraph START box supplies the paragraph-initial regime.

For A-2, a 600-tablet main shelf plus eight Paragraph START tablets reaches a practical point of diminishing returns. The fitted model closely reproduces several observed word-length and n -gram statistics. Across 5,000 simulation runs, 55.00% of generated tokens matched complete forms attested in A-2 itself, while an additional 22.05% matched forms attested elsewhere in the manuscript, giving a total manuscript attestation rate of 77.05%. This recurrence was obtained without storing any complete words or applying a whole-word rejection filter. Matched low-order null models show, however, that this high complete-word recurrence is largely a consequence of local transition structure rather than a shelf-specific effect. The more distinctive contribution of the reconstruction is its material implementation of these constraints while

maintaining a substantially narrower VM-like word-length distribution and greater lexical diversity than the corresponding unconstrained Markov generators.

These results do not establish the historical existence of the reconstructed apparatus. They do establish that a relatively simple analogue mechanism, feasible in principle without modern computation, is sufficient to reproduce a substantial subset of the statistical phenomena that motivated the reconstruction. The next stage is therefore not to treat the shelf as a solved historical identification, but to test whether the same architecture can be externally validated across additional bifolia, A/B regimes, paragraph phases, and physical layout constraints with progressively fewer fitted components.

A Reproducible Specification of the A-2 Shelf Reconstruction

A.1 Purpose and scope

This appendix specifies the fitted A-2 reconstruction in sufficient detail to distinguish the fixed mechanical architecture, the quantities estimated from A-2, and the technical completions introduced for simulation. The reconstruction uses only local character-transition events and stopping behavior. Complete Voynich word forms are neither stored in the apparatus nor used as an accept-reject criterion during generation.

The A-2 fitting material consists of the four surfaces f2r, f2v, f7r, and f7v. Under the filtering and character-segmentation conventions used in the main analysis, these surfaces contain 297 clean word tokens, 195 observed word types, and 155 hapax legomena.

The reported reconstruction is an in-sample fit of this material. It is not an externally trained predictor of A-2.

A.2 Fixed state space

The main shelf is a 24×24 array of open compartments. Both axes contain the same 24 states:

$$\{\text{SPACE}\} \cup \Sigma_{23},$$

where Σ_{23} is the working inventory of twenty-three VM character types used in the main analysis.

A compartment

$$[a, b]$$

stores the current two-state context. The compartment coordinate therefore carries the contextual information; an individual tablet need not repeat a or b .

Each tablet carries exactly two pieces of information:

$$c[n],$$

where c is the next VM character to be written and n is a stopping number.

Multiple tablets carrying the same $c[n]$ may be placed in the same compartment. Their multiplicity implements the relative weight of that output.

There are no separate END tablets and no END compartments.

A.3 State update and stopping rule

Suppose that the current state is

$$[a, b]$$

and that the selected tablet is $c[n]$. The operator first writes c . If the resulting word length is k , the stopping rule is

$$n \leq k \implies \text{terminate the word.}$$

If $n > k$, generation continues and the active state becomes

$$[b, c].$$

Thus ordinary continuation is

$$[a, b] \rightarrow c[n] \rightarrow [b, c].$$

If the tablet terminates the word, a SPACE is written and the state becomes

$$[c, \text{SPACE}].$$

The next word therefore begins through the sequence

$$[c, \text{SPACE}] \rightarrow d \rightarrow [\text{SPACE}, d] \rightarrow e \rightarrow [d, e].$$

This preserves the identity of the final character of the preceding word while the first character of the next word is selected.

The special state

$$[\text{SPACE}, \text{SPACE}]$$

is a neutral restart state. It suppresses immediate word-final-to-word-initial memory and is distinct from the paragraph-start mechanism.

A.4 Event classes used in fitting

The A-2 observations were separated into four mechanically distinct classes.

INTERNAL events.

These are ordinary word-internal transitions,

$$[a, b] \rightarrow c,$$

where both a and b are VM characters.

SECOND events.

These determine the second character of a word,

$$[\text{SPACE}, b] \rightarrow c.$$

BOUNDARY events.

These determine the first character of the next ordinary word while retaining the previous word-final character,

$$[b, \text{SPACE}] \rightarrow c.$$

Neutral restart events.

The state

$$[\text{SPACE}, \text{SPACE}]$$

was fitted separately as the neutral restart state.

For every state s and possible output c , the A-2 data provide an observed local count

$$N_s(c).$$

Within each event class, these counts define the empirical relative output weights. The fitted physical apparatus replaces those continuous relative weights with integer tablet multiplicities under a fixed total tablet budget.

Stopping numbers were fitted simultaneously from the observed stopping and continuation behavior at different word depths, subject to the single stopping rule specified above.

No term in this fitting procedure depends on whether the complete word ultimately produced by a sequence of transitions is an attested Voynich word.

A.5 Paragraph beginnings

Paragraph beginnings are deliberately excluded from the ordinary shelf optimization.

A-2 contains eight paragraph beginnings. Their observed initial-character distribution is

$$k : 6, \quad f : 1, \quad p : 1.$$

The A-2 Paragraph START box therefore contains exactly eight tablets:

$$k[8] \times 6, \quad f[8] \times 1, \quad p[8] \times 1.$$

At the beginning of a paragraph, one tablet is drawn from this box and its character c is written. The operator then enters the main shelf at

$$[\text{SPACE}, c].$$

The Paragraph START box is not used again until the next paragraph begins.

Consequently, the 100% paragraph-initial gallows rate reproduced by the A-2 simulation is a fitted property of this separate eight-tablet inventory and is not an independent prediction of the main shelf.

A.6 Selected inventory size

The same A-2 architecture was reconstructed with main-shelf budgets of

475, 500, 525, 550, 575, 600, 625, 650

tablets. The eight Paragraph START tablets were held fixed.

For each budget, A-2-sized generated samples were evaluated using multiple diagnostics rather than a single target statistic. These included:

- number of generated word types;
- number of hapax legomena;
- mean word length;
- dispersion of word length;
- frequency of very short words;
- frequency of words of eight or more characters;
- bigram Jensen–Shannon divergence;
- trigram Jensen–Shannon divergence; and
- word-boundary Jensen–Shannon divergence.

Increasing the main-shelf inventory from 475 to approximately 600 tablets produced substantial improvement in both lexical diversity and low-order transition fit. Increasing the inventory further to 625 or 650 tablets produced little additional reduction in trigram divergence and did not remove the remaining hapax deficit.

The value 600 was therefore selected as a practical elbow of the inventory–fit curve. It is not treated as a uniquely optimal number or as a historically privileged quantity.

A.7 Inventory of the selected reconstruction

The selected A-2 reconstruction contains 600 main-shelf tablets and eight Paragraph START tablets, for a total of

608 tablets

.

The main-shelf inventory is partitioned as follows.

The fourteen safety tablets occupy otherwise unobserved boundary states. Each is a minimal one-tablet completion introduced solely to prevent a finite simulation from entering an unrecoverable state. These tablets are not counted as empirical evidence that the corresponding A-2 transition was historically present.

The exact compartment, output character, stopping number, and multiplicity of every tablet in the fitted 600-tablet main-shelf configuration should be supplied in machine-readable form as Supplementary Table A1. For each distinct tablet type, that table should contain at least

Table 5 Tablet inventory of the selected A-2 reconstruction.

Tablet class	Number
INTERNAL	294
SECOND [SPACE, x]	146
BOUNDARY [x , SPACE]	111
Neutral restart [SPACE, SPACE]	35
Safety completion	14
Main-shelf total	600
Paragraph START	8
Grand total	608

(state₁, state₂, output character, stopping number, multiplicity, event class, safety flag).

The machine-readable inventory, rather than a regenerated approximation obtained by re-rounding the empirical frequencies, defines the exact configuration used for the simulations reported in the paper.

A.8 Required machine-readable inventory

For reproducibility, Supplementary Table A1 should use one row per distinct compartment–tablet combination, with the following fields.

Table 6 Required fields for Supplementary Table A1.

Field	Meaning
<code>state_1</code>	First coordinate of the active compartment
<code>state_2</code>	Second coordinate of the active compartment
<code>output</code>	Character written when the tablet is selected
<code>stop_number</code>	Integer stopping threshold n
<code>multiplicity</code>	Number of identical physical tablets
<code>class</code>	INTERNAL, SECOND, BOUNDARY, RESTART, or START
<code>safety</code>	1 for technical safety completion, otherwise 0

The sum of `multiplicity` over the main-shelf rows must equal

$$600,$$

and the sum over the Paragraph START rows must equal

$$8.$$

The class totals must reproduce

$$294 + 146 + 111 + 35 + 14 = 600.$$

The Paragraph START rows must reproduce

$$6 + 1 + 1 = 8.$$

Providing this table avoids ambiguity concerning integer rounding, low-frequency branches, and the precise placement of the fourteen safety completions.

A.9 Simulation procedure

A single simulated output is generated by the following mechanical procedure.

1. At a paragraph beginning, draw one tablet from Paragraph START.
2. Write its character c .
3. Enter the main shelf at $[\text{SPACE}, c]$.
4. Draw one tablet uniformly from the physical tablets present in the active compartment.
5. Write the character printed on the tablet.
6. Let k be the current number of VM characters in the word.
7. If the printed stopping number n satisfies $n \leq k$, write SPACE and enter $[c, \text{SPACE}]$, where c is the character just written.
8. Otherwise update the active two-character state in the ordinary way.
9. At an ordinary word boundary, preserve the previous final character through $[c, \text{SPACE}]$.
10. At a modeled neutral restart position, replace the current state by $[\text{SPACE}, \text{SPACE}]$.
11. Use Paragraph START only at the beginning of a paragraph, never at an ordinary line break or ordinary word boundary.
12. Continue until the required A-2-sized output has been produced.

The weak front-bias mechanism discussed in the main text is not used in the static A-2 fit assessment and must therefore be disabled when reproducing the reported fit statistics and complete-word recurrence results.

The exact paragraph and neutral-restart schedule used in the reported A-2 simulations should be supplied together with Supplementary Table A1, so that the simulation can be reproduced without inferring structural boundaries from the generated text.

A.10 Reproduction checks

An implementation of the published A-2 configuration should first verify the following invariant quantities before any statistical comparison is made:

$$N_{\text{main}} = 600,$$

$$N_{\text{START}} = 8,$$

$$N_{\text{total}} = 608,$$

and

$$N_{\text{safety}} = 14.$$

The Paragraph START distribution must be

$$k[8] \times 6, \quad f[8] \times 1, \quad p[8] \times 1.$$

No whole-word dictionary, whole-word probability table, or rejection filter may be consulted during generation.

For the simulation-based fit assessment, each run contains 297 generated word tokens. The published 1,000-run assessment gave the following means.

Table 7 Reproduction targets for the selected A-2 configuration.

Measure	A-2 observed	Simulation mean
Word types	195	188.54
Hapax legomena	155	136.17
Mean word length	3.9865	4.0186
Word-length SD	1.5763	1.6186
One-character words	4.71%	4.46%
Words of eight or more characters	1.35%	1.17%
Bigram JSD	—	0.03892
Trigram JSD	—	0.10608
Word-boundary JSD	—	0.11462
Paragraph-start gallows	100.00%	100.00%
Neutral-restart gallows	2.86%	2.77%

These values are reproduction targets for the specified fitted configuration, not independent predictive benchmarks.

A.11 Complete-word attestation check

A separate set of 5,000 independent A-2-sized runs was used to examine complete-word recurrence.

Generated tokens were classified hierarchically into four mutually exclusive categories:

1. attested in A-2;
2. absent from A-2 but attested elsewhere in Herbal A;
3. absent from Herbal A but attested elsewhere in the strict whole-manuscript comparison vocabulary; or
4. unattested in that comparison vocabulary.

The resulting mean token proportions were

$$55.00\%, \quad 17.52\%, \quad 4.53\%, \quad 22.95\%,$$

respectively.

Hence

$$55.00 + 17.52 + 4.53 = \boxed{77.05\%}$$

of generated tokens coincided with complete word forms attested somewhere in the manuscript.

Because A-2 itself was used to fit the tablet configuration, the 55.00% A-2 component is a within-sample recurrence measure. The quantity more relevant to generation beyond the fitting material is

$$17.52 + 4.53 = \boxed{22.05 \text{ percentage points}},$$

representing generated forms absent from A-2 but independently attested elsewhere in the manuscript.

This comparison is performed only after generation has been completed. Attestation status is never consulted by the apparatus during word construction.

A.12 Reproducibility files

A complete reproducibility package for the A-2 result should therefore contain:

- the filtered A-2 input transcription used for fitting;
- the exact 600-tablet main-shelf inventory;
- the eight Paragraph START tablets;
- identification of the fourteen safety tablets;
- the paragraph and neutral-restart schedule;
- the code implementing the state-update and stopping rules;
- the random-number seed or seeds used for any exactly reproduced run;
- the 1,000-run fit-assessment output;
- the 5,000-run complete-word-attestation output; and
- the three comparison vocabularies used for A-2, Herbal A, and the strict whole-manuscript attestation test.

With these materials, the reported simulation does not require reconstruction of any undocumented fitting decision: the published tablet inventory itself is the definitive integer realization of the fitted A-2 apparatus.

B Exact A-2 600-tablet main-shelf inventory

Each line gives one distinct compartment-tablet combination. Multiplicity is the number of identical physical tablets.

Note. The **class** field is an analytical annotation and is not written on the physical tablets. **INTERNAL** denotes ordinary within-word transitions, **SECOND** the selection of a word's second character from a state of the form [SPACE, *b*], **BOUNDARY** the selection of the next word's first character from [*b*, SPACE], **RESTART** a neutral restart from [SPACE, SPACE], and **START** the separate paragraph-initial selection. The **safety** field is likewise an analytical flag rather than part of the tablet inscription.

RESTART

[SPACE, SPACE] → q[8] × 5; class=RESTART; safety=0
 [SPACE, SPACE] → ch[8] × 5; class=RESTART; safety=0
 [SPACE, SPACE] → sh[8] × 4; class=RESTART; safety=0
 [SPACE, SPACE] → k[8] × 1; class=RESTART; safety=0
 [SPACE, SPACE] → o[8] × 7; class=RESTART; safety=0

[SPACE, SPACE] $\rightarrow$ d[8] $\times$ 9; class=RESTART; safety=0
 [SPACE, SPACE] $\rightarrow$ s[1] $\times$ 2; class=RESTART; safety=0
 [SPACE, SPACE] $\rightarrow$ s[8] $\times$ 1; class=RESTART; safety=0
 [SPACE, SPACE] $\rightarrow$ y[8] $\times$ 1; class=RESTART; safety=0

SECOND

[SPACE, q] $\rightarrow$ k[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, q] $\rightarrow$ o[8] $\times$ 10; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ cth[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ ckh[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ k[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ o[2] $\times$ 1; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ o[8] $\times$ 14; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ t[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ a[8] $\times$ 3; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ e[8] $\times$ 6; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ y[2] $\times$ 2; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ y[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, ch] $\rightarrow$ r[1] $\times$ 1; class=SECOND; safety=0
 [SPACE, cph] $\rightarrow$ e[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, cph] $\rightarrow$ y[1] $\times$ 1; class=SECOND; safety=0
 [SPACE, cth] $\rightarrow$ o[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, cth] $\rightarrow$ d[1] $\times$ 1; class=SECOND; safety=0
 [SPACE, cth] $\rightarrow$ e[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, cth] $\rightarrow$ y[1] $\times$ 2; class=SECOND; safety=0
 [SPACE, p] $\rightarrow$ ch[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, p] $\rightarrow$ o[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, sh] $\rightarrow$ ckh[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, sh] $\rightarrow$ o[2] $\times$ 3; class=SECOND; safety=0
 [SPACE, sh] $\rightarrow$ o[8] $\times$ 6; class=SECOND; safety=0
 [SPACE, sh] $\rightarrow$ t[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, sh] $\rightarrow$ a[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, sh] $\rightarrow$ e[8] $\times$ 5; class=SECOND; safety=0
 [SPACE, sh] $\rightarrow$ y[1] $\times$ 1; class=SECOND; safety=0
 [SPACE, ckh] $\rightarrow$ o[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, ckh] $\rightarrow$ y[1] $\times$ 1; class=SECOND; safety=0
 [SPACE, f] $\rightarrow$ ch[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, f] $\rightarrow$ o[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, k] $\rightarrow$ ch[8] $\times$ 3; class=SECOND; safety=0
 [SPACE, k] $\rightarrow$ sh[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, k] $\rightarrow$ o[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, k] $\rightarrow$ e[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, k] $\rightarrow$ y[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, o] $\rightarrow$ k[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, o] $\rightarrow$ t[8] $\times$ 3; class=SECOND; safety=0
 [SPACE, o] $\rightarrow$ a[8] $\times$ 1; class=SECOND; safety=0

[SPACE, o] $\rightarrow$ d[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, o] $\rightarrow$ e[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, o] $\rightarrow$ l[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, t] $\rightarrow$ ch[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, t] $\rightarrow$ sh[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, t] $\rightarrow$ e[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, t] $\rightarrow$ y[1] $\times$ 1; class=SECOND; safety=0
 [SPACE, a] $\rightarrow$ i[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, a] $\rightarrow$ l[2] $\times$ 1; class=SECOND; safety=0
 [SPACE, a] $\rightarrow$ n[1] $\times$ 1; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ ch[8] $\times$ 3; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ sh[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ k[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ o[8] $\times$ 6; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ a[8] $\times$ 15; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ e[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ l[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ y[2] $\times$ 2; class=SECOND; safety=0
 [SPACE, d] $\rightarrow$ y[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, s] $\rightarrow$ o[2] $\times$ 1; class=SECOND; safety=0
 [SPACE, s] $\rightarrow$ a[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, l] $\rightarrow$ sh[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, l] $\rightarrow$ o[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, y] $\rightarrow$ ch[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, y] $\rightarrow$ p[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, y] $\rightarrow$ k[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, y] $\rightarrow$ o[8] $\times$ 1; class=SECOND; safety=0
 [SPACE, y] $\rightarrow$ t[8] $\times$ 2; class=SECOND; safety=0
 [SPACE, y] $\rightarrow$ d[8] $\times$ 1; class=SECOND; safety=0
BOUNDARY
 [q, SPACE] $\rightarrow$ ch[8] $\times$ 1; class=BOUNDARY; safety=1
INTERNAL
 [q, k] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ p[7] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ k[4] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ k[5] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ k[7] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ t[4] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ t[5] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ t[7] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ t[8] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ s[7] $\times$ 1; class=INTERNAL; safety=0
 [q, o] $\rightarrow$ y[1] $\times$ 1; class=INTERNAL; safety=0
BOUNDARY

[cfh, SPACE] $\rightarrow$ ch[8] $\times$ 1; class=BOUNDARY; safety=1

[ch, SPACE] $\rightarrow$ ch[8] $\times$ 1; class=BOUNDARY; safety=1

INTERNAL

[ch, cth] $\rightarrow$ y[1] $\times$ 1; class=INTERNAL; safety=0

[ch, ckh] $\rightarrow$ y[1] $\times$ 1; class=INTERNAL; safety=0

[ch, k] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0

[ch, k] $\rightarrow$ a[7] $\times$ 1; class=INTERNAL; safety=0

[ch, k] $\rightarrow$ e[7] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ ch[5] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ k[4] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ k[7] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ t[4] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ t[7] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ a[7] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ d[4] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ d[7] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ d[8] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ s[7] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ i[8] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ l[2] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ l[3] $\times$ 2; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ l[7] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ l[8] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ r[1] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ r[3] $\times$ 2; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ r[7] $\times$ 1; class=INTERNAL; safety=0

[ch, o] $\rightarrow$ r[8] $\times$ 1; class=INTERNAL; safety=0

[ch, t] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0

[ch, t] $\rightarrow$ y[1] $\times$ 1; class=INTERNAL; safety=0

[ch, a] $\rightarrow$ i[7] $\times$ 1; class=INTERNAL; safety=0

[ch, a] $\rightarrow$ i[8] $\times$ 1; class=INTERNAL; safety=0

[ch, a] $\rightarrow$ l[3] $\times$ 1; class=INTERNAL; safety=0

[ch, a] $\rightarrow$ r[3] $\times$ 1; class=INTERNAL; safety=0

[ch, a] $\rightarrow$ n[1] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ o[5] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ o[8] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ t[7] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ a[7] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ e[6] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ e[7] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ e[8] $\times$ 1; class=INTERNAL; safety=0

[ch, e] $\rightarrow$ y[1] $\times$ 4; class=INTERNAL; safety=0

[ch, y] $\rightarrow$ k[7] $\times$ 1; class=INTERNAL; safety=0

BOUNDARY

$[cph, SPACE] \rightarrow ch[8] \times 1; \text{class=BOUNDARY; safety=1}$
INTERNAL
 $[cph, e] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[cth, SPACE] \rightarrow ch[8] \times 1; \text{class=BOUNDARY; safety=1}$
INTERNAL
 $[cth, o] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[cth, o] \rightarrow l[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[cth, o] \rightarrow r[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[cth, e] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[p, SPACE] \rightarrow ch[8] \times 1; \text{class=BOUNDARY; safety=1}$
INTERNAL
 $[p, ch] \rightarrow o[5] \times 1; \text{class=INTERNAL; safety=0}$
 $[p, ch] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[p, ch] \rightarrow a[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[p, ch] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[p, ch] \rightarrow y[2] \times 1; \text{class=INTERNAL; safety=0}$
 $[p, o] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[sh, SPACE] \rightarrow o[8] \times 1; \text{class=BOUNDARY; safety=0}$
INTERNAL
 $[sh, ckh] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow p[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow d[4] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow d[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow l[3] \times 2; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow l[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow l[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow r[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, o] \rightarrow r[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, t] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, a] \rightarrow i[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, a] \rightarrow n[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, e] \rightarrow k[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, e] \rightarrow o[5] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, e] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, e] \rightarrow e[6] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, e] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, e] \rightarrow e[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[sh, e] \rightarrow y[1] \times 2; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[ckh, SPACE] \rightarrow ch[8] \times 1; \text{class=BOUNDARY; safety=1}$

INTERNAL
 $[ckh, o] \rightarrow l[3] \times 1$; class=INTERNAL; safety=0
 $[ckh, o] \rightarrow r[3] \times 1$; class=INTERNAL; safety=0
 BOUNDARY
 $[f, SPACE] \rightarrow ch[8] \times 1$; class=BOUNDARY; safety=1
 INTERNAL
 $[f, ch] \rightarrow o[7] \times 1$; class=INTERNAL; safety=0
 $[f, o] \rightarrow d[7] \times 1$; class=INTERNAL; safety=0
 BOUNDARY
 $[k, SPACE] \rightarrow ch[8] \times 1$; class=BOUNDARY; safety=1
 INTERNAL
 $[k, ch] \rightarrow o[5] \times 1$; class=INTERNAL; safety=0
 $[k, ch] \rightarrow o[7] \times 1$; class=INTERNAL; safety=0
 $[k, ch] \rightarrow e[7] \times 1$; class=INTERNAL; safety=0
 $[k, ch] \rightarrow y[2] \times 1$; class=INTERNAL; safety=0
 $[k, ch] \rightarrow y[3] \times 1$; class=INTERNAL; safety=0
 $[k, sh] \rightarrow o[7] \times 1$; class=INTERNAL; safety=0
 $[k, sh] \rightarrow y[1] \times 1$; class=INTERNAL; safety=0
 $[k, o] \rightarrow o[7] \times 1$; class=INTERNAL; safety=0
 $[k, o] \rightarrow d[7] \times 1$; class=INTERNAL; safety=0
 $[k, o] \rightarrow i[8] \times 1$; class=INTERNAL; safety=0
 $[k, o] \rightarrow l[3] \times 1$; class=INTERNAL; safety=0
 $[k, o] \rightarrow l[7] \times 1$; class=INTERNAL; safety=0
 $[k, o] \rightarrow l[8] \times 1$; class=INTERNAL; safety=0
 $[k, a] \rightarrow i[8] \times 1$; class=INTERNAL; safety=0
 $[k, a] \rightarrow r[3] \times 1$; class=INTERNAL; safety=0
 $[k, e] \rightarrow ch[5] \times 1$; class=INTERNAL; safety=0
 $[k, e] \rightarrow o[5] \times 1$; class=INTERNAL; safety=0
 $[k, e] \rightarrow o[7] \times 1$; class=INTERNAL; safety=0
 $[k, e] \rightarrow e[6] \times 1$; class=INTERNAL; safety=0
 $[k, e] \rightarrow e[7] \times 1$; class=INTERNAL; safety=0
 $[k, e] \rightarrow y[1] \times 1$; class=INTERNAL; safety=0
 $[k, y] \rightarrow d[7] \times 1$; class=INTERNAL; safety=0
 BOUNDARY
 $[o, SPACE] \rightarrow ch[8] \times 3$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow p[8] \times 1$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow sh[8] \times 1$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow k[8] \times 3$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow t[8] \times 1$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow d[8] \times 1$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow l[1] \times 1$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow l[8] \times 1$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow y[8] \times 1$; class=BOUNDARY; safety=0
 $[o, SPACE] \rightarrow r[1] \times 2$; class=BOUNDARY; safety=0

INTERNAL

[o, ch] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0
[o, ch] $\rightarrow$ e[7] $\times$ 1; class=INTERNAL; safety=0
[o, ch] $\rightarrow$ y[2] $\times$ 1; class=INTERNAL; safety=0
[o, ch] $\rightarrow$ y[3] $\times$ 1; class=INTERNAL; safety=0
[o, p] $\rightarrow$ ch[5] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ ch[5] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ ch[7] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ ch[8] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ sh[7] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ o[5] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ a[7] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ e[6] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ e[7] $\times$ 1; class=INTERNAL; safety=0
[o, k] $\rightarrow$ y[2] $\times$ 1; class=INTERNAL; safety=0
[o, o] $\rightarrow$ s[7] $\times$ 1; class=INTERNAL; safety=0
[o, o] $\rightarrow$ i[8] $\times$ 1; class=INTERNAL; safety=0
[o, t] $\rightarrow$ ch[5] $\times$ 1; class=INTERNAL; safety=0
[o, t] $\rightarrow$ ch[7] $\times$ 1; class=INTERNAL; safety=0
[o, t] $\rightarrow$ ch[8] $\times$ 1; class=INTERNAL; safety=0
[o, t] $\rightarrow$ sh[7] $\times$ 1; class=INTERNAL; safety=0
[o, t] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0
[o, t] $\rightarrow$ a[7] $\times$ 1; class=INTERNAL; safety=0
[o, t] $\rightarrow$ e[7] $\times$ 1; class=INTERNAL; safety=0
[o, t] $\rightarrow$ y[1] $\times$ 1; class=INTERNAL; safety=0
[o, a] $\rightarrow$ i[8] $\times$ 1; class=INTERNAL; safety=0
[o, d] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0
[o, d] $\rightarrow$ a[5] $\times$ 1; class=INTERNAL; safety=0
[o, d] $\rightarrow$ a[6] $\times$ 1; class=INTERNAL; safety=0
[o, d] $\rightarrow$ a[7] $\times$ 1; class=INTERNAL; safety=0
[o, d] $\rightarrow$ a[8] $\times$ 1; class=INTERNAL; safety=0
[o, d] $\rightarrow$ y[2] $\times$ 2; class=INTERNAL; safety=0
[o, d] $\rightarrow$ y[3] $\times$ 1; class=INTERNAL; safety=0
[o, e] $\rightarrow$ e[7] $\times$ 1; class=INTERNAL; safety=0
[o, s] $\rightarrow$ ch[5] $\times$ 1; class=INTERNAL; safety=0
[o, i] $\rightarrow$ sh[7] $\times$ 1; class=INTERNAL; safety=0
[o, i] $\rightarrow$ i[7] $\times$ 1; class=INTERNAL; safety=0
[o, i] $\rightarrow$ i[8] $\times$ 1; class=INTERNAL; safety=0
[o, l] $\rightarrow$ sh[7] $\times$ 1; class=INTERNAL; safety=0
[o, l] $\rightarrow$ o[5] $\times$ 1; class=INTERNAL; safety=0
[o, l] $\rightarrow$ o[7] $\times$ 1; class=INTERNAL; safety=0
[o, l] $\rightarrow$ a[7] $\times$ 1; class=INTERNAL; safety=0
[o, l] $\rightarrow$ d[4] $\times$ 1; class=INTERNAL; safety=0
[o, l] $\rightarrow$ d[7] $\times$ 1; class=INTERNAL; safety=0

$[o, l] \rightarrow s[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[o, l] \rightarrow s[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[o, l] \rightarrow y[2] \times 1; \text{class=INTERNAL; safety=0}$
 $[o, r] \rightarrow \text{ch}[5] \times 1; \text{class=INTERNAL; safety=0}$
 $[o, r] \rightarrow \text{ch}[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[o, r] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[t, \text{SPACE}] \rightarrow \text{ch}[8] \times 1; \text{class=BOUNDARY; safety=1}$
INTERNAL
 $[t, \text{ch}] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, \text{ch}] \rightarrow a[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, \text{ch}] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, \text{ch}] \rightarrow y[2] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, \text{sh}] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, o] \rightarrow \text{ch}[5] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, o] \rightarrow d[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, o] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, o] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, a] \rightarrow i[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, e] \rightarrow e[6] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, e] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[t, e] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[a, \text{SPACE}] \rightarrow \text{ch}[8] \times 1; \text{class=BOUNDARY; safety=1}$
INTERNAL
 $[a, i] \rightarrow i[5] \times 1; \text{class=INTERNAL; safety=0}$
 $[a, i] \rightarrow i[7] \times 2; \text{class=INTERNAL; safety=0}$
 $[a, i] \rightarrow i[8] \times 3; \text{class=INTERNAL; safety=0}$
 $[a, i] \rightarrow r[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[a, i] \rightarrow n[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[a, i] \rightarrow n[5] \times 1; \text{class=INTERNAL; safety=0}$
 $[a, l] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[a, l] \rightarrow d[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[a, l] \rightarrow s[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[a, r] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[d, \text{SPACE}] \rightarrow \text{sh}[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[d, \text{SPACE}] \rightarrow d[8] \times 2; \text{class=BOUNDARY; safety=0}$
INTERNAL
 $[d, \text{ch}] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, \text{ch}] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, \text{ch}] \rightarrow y[2] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, \text{sh}] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, k] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, o] \rightarrow d[7] \times 1; \text{class=INTERNAL; safety=0}$

$[d, o] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, o] \rightarrow l[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, o] \rightarrow r[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, o] \rightarrow r[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, a] \rightarrow i[5] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, a] \rightarrow i[7] \times 2; \text{class=INTERNAL; safety=0}$
 $[d, a] \rightarrow i[8] \times 2; \text{class=INTERNAL; safety=0}$
 $[d, a] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, a] \rightarrow n[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, e] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, e] \rightarrow e[6] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, e] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, l] \rightarrow s[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[d, y] \rightarrow k[7] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[e, \text{SPACE}] \rightarrow \text{ch}[8] \times 1; \text{class=BOUNDARY; safety=0}$
INTERNAL
 $[e, \text{ch}] \rightarrow y[2] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, k] \rightarrow y[2] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, o] \rightarrow d[4] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, o] \rightarrow d[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, o] \rightarrow s[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, o] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, o] \rightarrow l[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, o] \rightarrow r[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, t] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, a] \rightarrow i[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, a] \rightarrow r[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, e] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, e] \rightarrow a[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, e] \rightarrow e[6] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, e] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, e] \rightarrow s[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, e] \rightarrow s[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[e, e] \rightarrow y[1] \times 3; \text{class=INTERNAL; safety=0}$
 $[e, s] \rightarrow e[1] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[s, \text{SPACE}] \rightarrow q[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[s, \text{SPACE}] \rightarrow \text{ch}[8] \times 3; \text{class=BOUNDARY; safety=0}$
 $[s, \text{SPACE}] \rightarrow \text{sh}[8] \times 3; \text{class=BOUNDARY; safety=0}$
 $[s, \text{SPACE}] \rightarrow o[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[s, \text{SPACE}] \rightarrow d[8] \times 1; \text{class=BOUNDARY; safety=0}$
INTERNAL
 $[s, \text{ch}] \rightarrow y[2] \times 1; \text{class=INTERNAL; safety=0}$
 $[s, o] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$

$[s, o] \rightarrow r[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[s, a] \rightarrow i[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[s, a] \rightarrow i[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[s, a] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[i, \text{SPACE}] \rightarrow ch[8] \times 1; \text{class=BOUNDARY; safety=1}$
INTERNAL
 $[i, sh] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, s] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow s[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow i[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow l[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow r[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow n[2] \times 2; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow n[3] \times 5; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow n[5] \times 2; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow n[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, i] \rightarrow n[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, l] \rightarrow d[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[i, n] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
BOUNDARY
 $[l, \text{SPACE}] \rightarrow q[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow ch[8] \times 4; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow cph[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow cth[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow sh[8] \times 2; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow f[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow k[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow o[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow d[8] \times 5; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow s[8] \times 1; \text{class=BOUNDARY; safety=0}$
 $[l, \text{SPACE}] \rightarrow y[8] \times 1; \text{class=BOUNDARY; safety=0}$
INTERNAL
 $[l, sh] \rightarrow o[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, sh] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, sh] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, o] \rightarrow ch[5] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, o] \rightarrow d[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, o] \rightarrow e[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, o] \rightarrow r[3] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, a] \rightarrow i[8] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, d] \rightarrow y[2] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, s] \rightarrow a[7] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, s] \rightarrow y[1] \times 1; \text{class=INTERNAL; safety=0}$
 $[l, y] \rightarrow sh[7] \times 1; \text{class=INTERNAL; safety=0}$

[y, SPACE] $\rightarrow$ q[8] $\times$ 2; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ ch[8] $\times$ 4; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ cph[8] $\times$ 1; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ cth[8] $\times$ 1; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ sh[8] $\times$ 2; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ ckh[8] $\times$ 1; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ k[8] $\times$ 2; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ o[1] $\times$ 1; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ o[8] $\times$ 1; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ t[8] $\times$ 1; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ d[8] $\times$ 4; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ s[8] $\times$ 1; class=BOUNDARY; safety=0
[y, SPACE] $\rightarrow$ y[8] $\times$ 2; class=BOUNDARY; safety=0

```

[y, ch] → a[7] × 1; class=INTERNAL; safety=0
[y, ch] → e[7] × 1; class=INTERNAL; safety=0
[y, p] → ch[5] × 1; class=INTERNAL; safety=0
[y, sh] → y[1] × 1; class=INTERNAL; safety=0
[y, k] → ch[5] × 1; class=INTERNAL; safety=0
[y, k] → sh[7] × 1; class=INTERNAL; safety=0
[y, k] → o[7] × 1; class=INTERNAL; safety=0
[y, k] → e[7] × 1; class=INTERNAL; safety=0
[y, k] → y[2] × 1; class=INTERNAL; safety=0
[y, k] → y[3] × 1; class=INTERNAL; safety=0
[y, o] → r[3] × 1; class=INTERNAL; safety=0
[y, t] → ch[5] × 1; class=INTERNAL; safety=0
[y, t] → ch[7] × 1; class=INTERNAL; safety=0
[y, t] → o[7] × 1; class=INTERNAL; safety=0
[y, d] → a[5] × 1; class=INTERNAL; safety=0
[y, d] → a[7] × 1; class=INTERNAL; safety=0
[y, d] → y[2] × 1; class=INTERNAL; safety=0

```

[r, SPACE]	→ q[8] × 1; class=BOUNDARY; safety=0
[r, SPACE]	→ ch[8] × 6; class=BOUNDARY; safety=0
[r, SPACE]	→ cth[8] × 2; class=BOUNDARY; safety=0
[r, SPACE]	→ sh[8] × 3; class=BOUNDARY; safety=0
[r, SPACE]	→ ckh[8] × 1; class=BOUNDARY; safety=0
[r, SPACE]	→ o[8] × 2; class=BOUNDARY; safety=0
[r, SPACE]	→ a[8] × 2; class=BOUNDARY; safety=0
[r, SPACE]	→ d[8] × 3; class=BOUNDARY; safety=0
[r, SPACE]	→ s[1] × 1; class=BOUNDARY; safety=0
[r, SPACE]	→ l[8] × 1; class=BOUNDARY; safety=0
[r, SPACE]	→ y[8] × 2; class=BOUNDARY; safety=0

$[r, ch] \rightarrow o[7] \times 1$; class=INTERNAL; safety=0
 $[r, ch] \rightarrow e[7] \times 1$; class=INTERNAL; safety=0
 $[r, o] \rightarrow ch[5] \times 1$; class=INTERNAL; safety=0
BOUNDARY
 $[m, SPACE] \rightarrow ch[8] \times 1$; class=BOUNDARY; safety=1
 $[n, SPACE] \rightarrow q[8] \times 1$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow ch[8] \times 3$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow cth[8] \times 1$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow sh[8] \times 3$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow ckh[8] \times 1$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow o[8] \times 2$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow t[8] \times 1$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow d[1] \times 1$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow d[8] \times 1$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow s[8] \times 1$; class=BOUNDARY; safety=0
 $[n, SPACE] \rightarrow y[8] \times 1$; class=BOUNDARY; safety=0
 $[g, SPACE] \rightarrow ch[8] \times 1$; class=BOUNDARY; safety=1

Main-shelf total: 600 tablets.

Class totals: INTERNAL 294; SECOND 146; BOUNDARY 125 (including 14 safety completions); RESTART 35.

Safety-completion total: 14 tablets.

Separate Paragraph START inventory (not included in the 600 main-shelf tablets):

$[PARAGRAPH START] \rightarrow k[8] \times 6$
 $[PARAGRAPH START] \rightarrow f[8] \times 1$
 $[PARAGRAPH START] \rightarrow p[8] \times 1$

Paragraph START total: 8 tablets; grand total: 608 tablets.

Declarations

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflicts of Interest

The author declares no conflict of interest.

Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data Availability

No empirical data was used in this study.

References

- [1] Bown, C.L., Lindemann, L.: The linguistics of the voynich manuscript. *Annual Review of Linguistics* **7**, 285–308 (2021) <https://doi.org/10.1146/annurev-linguistics-011619-030613>

- [2] Zandbergen, R.: ZL Transliteration File. Version 3b, IVTFF 2.0. Version dated 13 May 2025 (2025). <https://www.voynich.nu/data/ZL3b-n.txt>
- [3] Zandbergen, R.: Transliteration of the voynich MS text. In: Layfield, C., Abela, J. (eds.) Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022). CEUR Workshop Proceedings, vol. 3313. CEUR-WS.org, ??? (2022). <https://ceur-ws.org/Vol-3313/keynote1.pdf>
- [4] Currier, P.H.: Some important new statistical findings. In: D’Imperio, M.E. (ed.) New Research on the Voynich Manuscript: Proceedings of a Seminar. Privately printed, Washington, D.C. (1976). Seminar held 30 November 1976; later electronic transcription by Jacques Guy and Jim Reeds, distributed by René Zandbergen. https://www.voynich.nu/extra/curr_main.html
- [5] Davis, L.F.: How many glyphs and how many scribes? digital paleography and the voynich manuscript. Manuscript Studies: A Journal of the Schoenberg Institute for Manuscript Studies **5**(1), 164–180 (2020) <https://doi.org/10.1353/mns.2020.0011>
- [6] Farrugia, K., Layfield, C., Plas, L.: Demystifying the scribes behind the voynich manuscript using computational linguistic techniques. In: Layfield, C., Abela, J. (eds.) Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022). CEUR Workshop Proceedings, vol. 3313. CEUR-WS.org, ??? (2022). <https://ceur-ws.org/Vol-3313/paper5.pdf>
- [7] Bonner, A.: The Art and Logic of Ramon Llull: A User’s Guide. Studien und Texte zur Geistesgeschichte des Mittelalters, vol. 95. Brill, Leiden and Boston (2007). <https://doi.org/10.1163/ej.9789004163256.i-342>
- [8] Pelling, N.: The Curse of the Voynich: The Secret History of the World’s Most Mysterious Manuscript. Compelling Press, ??? (2006)